

3-M4 (LVM3-M4) [in the] Chandrayaan and Aditya L1 missions. L&T is involved in these missions, starting from the manufacturing of flight hardware to mission tracking,” states Arun Ramchandani, Executive Vice President and Head of Defence Business at L&T.

Then there are component suppliers like Pune-based Walchandnagar Industries, which has devoted a significant part of its resources towards servicing space services in the five decades of its association with Isro. Chirag Doshi, Managing Director & CEO, says, “Since our division’s humble beginnings in the early 1970s, we boast of 40 per cent of our resources, both men and machines, exclusively working towards India’s space programme.”

A second wave of private sector partnerships has been forged with the expansion of Isro’s space programme since the 1990s, with new suppliers like Ananth Technologies Ltd (ATL), Alpha Design Technologies Ltd (ADTL), Astra Microwave Products, Centum Electronics, Grintex India, and MISSAR Systems. Hyderabad-based ATL has been engaged in the production of diverse electronic and mechanical sub-systems for Isro’s launch vehicles, satellites, spacecraft payloads, and ground systems. Referring to the recent successful launch of the Aditya L1 mission, Subba Rao Pavuluri, ATL Founder, Chairman and Managing Director, observes, “This partnership represents a significant milestone for us, as we contribute our technical excellence and manufacturing to support India’s space exploration endeavours.”

In 2016, ADTL became the first Indian company to secure a contract for Assembly Integration and Testing (AIT) for the Indian Regional Navigation Satellite programme.

“The same year, Isro also awarded us the contract for the revival of the tele-education network in the Northeastern states by upgrading seven Ku-band hubs and about 325 Very Small Aperture Terminal (VSAT)-based classrooms,” mentions Sujatha Deepak, General Manager-Systems at ADTL. Ku-band satellites are the most popular for all forms of communications while VSAT is a small-sized earth station for receiving or transmitting data.

VENDORS NO MORE

But as Space Race 2.0 picks up pace, the companies are no longer seen just as vendors. The government would like them to play a larger role in building a robust spacetechnology ecosystem. This change in thinking has been prompted by the transformation of the global space industry. Four major trends, including the mass adoption of commercial satellite services mainly for providing internet access, commercial space stations, in-space resource extraction and exploitation, and weaponisation of space, especially, indicate pivotal ‘strategic turning points’ for the space sector, according to a report by the advisory Arthur D. Little. Experience gained from space activities can help the private sector increase the role of technology in other areas and boost the pace of the Indian space sector, bringing it in line with mature space agencies like US space agency Nasa, says TCE’s Singh. “Private sector involvement not only brings strong project management and planning skills to the project but also technical design and engineering skills,” he adds.

Following the successful lift-off of India’s first privately built Vikram-S rocket last November, the ambitions of several spacetechnology start-ups have also soared. With the



PAWAN GOENKA
CHAIRMAN, IN-SPACE

“Larger companies, which were primarily vendors to Isro, have now started warming up to the idea of investing in technology and making bigger bets. So many want to have access to making rockets”



SRIMATHY KESAN
FOUNDER AND CEO,
SPACE KIDZ INDIA

“We are living in the best of times. From a child to a researcher, everyone wants to go to space. But... there’s also a lot of other learnings in the process. That’s the reason you also have so many start-ups coming up”

rocket serving as the best metaphor for space exploration, these companies that are developing everything from launch vehicles to satellites to space applications see Kondapur (Telangana)-based Skyroot Aerospace’s achievement as putting their sector on a high growth trajectory. “India has a space ecosystem built over six decades, with a very